

E-COMMERCE CUSTOMER CHURN PREDICTION SYSTEM

Final Project Report

Internship Program

Innolift Ventures Internship Program

Submitted By

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Track: Machine Learning & Flask Development

Project Overview

The E-Commerce Customer Churn Prediction System is a Machine Learning-based web application designed to predict whether a customer is likely to discontinue using a company's services. The system helps businesses identify at-risk customers and take proactive retention measures.

The application combines Machine Learning, Flask Web Development, SQLite Database Management, and Data Analytics to provide real-time customer churn predictions along with risk assessment and customer retention recommendations.

Problem Statement

Customer churn is one of the major challenges faced by e-commerce businesses. Acquiring new customers is significantly more expensive than retaining existing customers.

The objective of this project is to:

- Predict customer churn probability.
- Identify high-risk customers.
- Provide retention recommendations.
- Help businesses improve customer retention strategies.
- Visualize customer risk distribution.

Project Objectives

- Develop a machine learning model for churn prediction.
- Build a complete Flask web application.
- Store customer records using SQLite.
- Display prediction results dynamically.
- Classify customers into risk categories.
- Generate retention recommendations.
- Provide an interactive dashboard for business insights.

Technology Stack

Frontend

- HTML5

- CSS3

Backend

- Python
- Flask

Database

- SQLite

Machine Learning

- Scikit-Learn
- Logistic Regression

Libraries Used

- Pandas
- NumPy
- Joblib

System Architecture

User Input



HTML Form



Flask Backend



Data Processing



Machine Learning Model (.pkl)



Prediction Engine



Risk Classification



SQLite Database Storage



Dashboard & Reports

Dataset Features

The model uses the following customer attributes:

Feature	Description
Customer ID	Unique customer identifier
Age	Customer age
Tenure Months	Duration of customer relationship
Average Order Value	Average customer spending
Total Orders	Total purchase count
Last Purchase Days Ago	Customer inactivity period
Support Tickets	Number of complaints/support requests
Gender	Customer gender
City	Customer location
Subscription Type	Customer subscription plan

Machine Learning Model

Model Used:

Logistic Regression

Model Workflow:

1. Data preprocessing
2. Feature engineering
3. Train-test split
4. Model training
5. Hyperparameter tuning
6. Model evaluation
7. Model serialization using Joblib
8. Deployment using Flask

Model File:

final_model.pkl

Key Features Implemented

Customer Churn Prediction

Predicts whether a customer is likely to churn.

Confidence Score

Displays model confidence percentage.

Risk Classification

Customers are categorized as:

- High Risk
- Medium Risk
- Low Risk

Risk Distribution Dashboard

Displays:

- High Risk Customers
- Medium Risk Customers
- Low Risk Customers
- Total Predictions

Customer Records Management

- Store predictions
- View records
- Delete records

Risk-Based Customer Filtering

Users can click risk categories and view customers belonging to that category.

Retention Recommendations

Provides actionable suggestions to retain customers.

Risk Factors Analysis

Displays the major reasons contributing to churn risk.

Application Modules

Home Page

- Displays project overview and risk distribution summary.

Prediction Page

- Accepts customer information and generates churn predictions.

Dashboard

- Provides business insights and customer statistics.

Customer Records

- Displays all customer prediction records.

Risk Customers Page

- Shows customers grouped by risk category.

About Page

- Provides project details and feature descriptions.

Database Design

Database: SQLite

Table Name: customers

Stored Information:

- Customer Details
- Prediction Result
- Confidence Score
- Risk Level

Project Workflow

- User enters customer information.
- Flask receives form data.

- Data is converted into model input format.
- Machine learning model generates prediction.
- Churn probability is calculated.
- Risk level is assigned.
- Risk factors are identified.
- Retention recommendations are generated.
- Data is stored in SQLite database.
- Results are displayed to the user.

Results Achieved

Successfully implemented:

Machine Learning Model Integration

Flask Backend Development

SQLite Database Integration

Dynamic Prediction System

Customer Risk Classification

Customer Retention Recommendations

Dashboard Analytics

End-to-End Workflow

Challenges Faced

Feature encoding for model deployment.

Flask route integration.

Database schema updates.

Dynamic risk classification implementation.

UI consistency across pages.

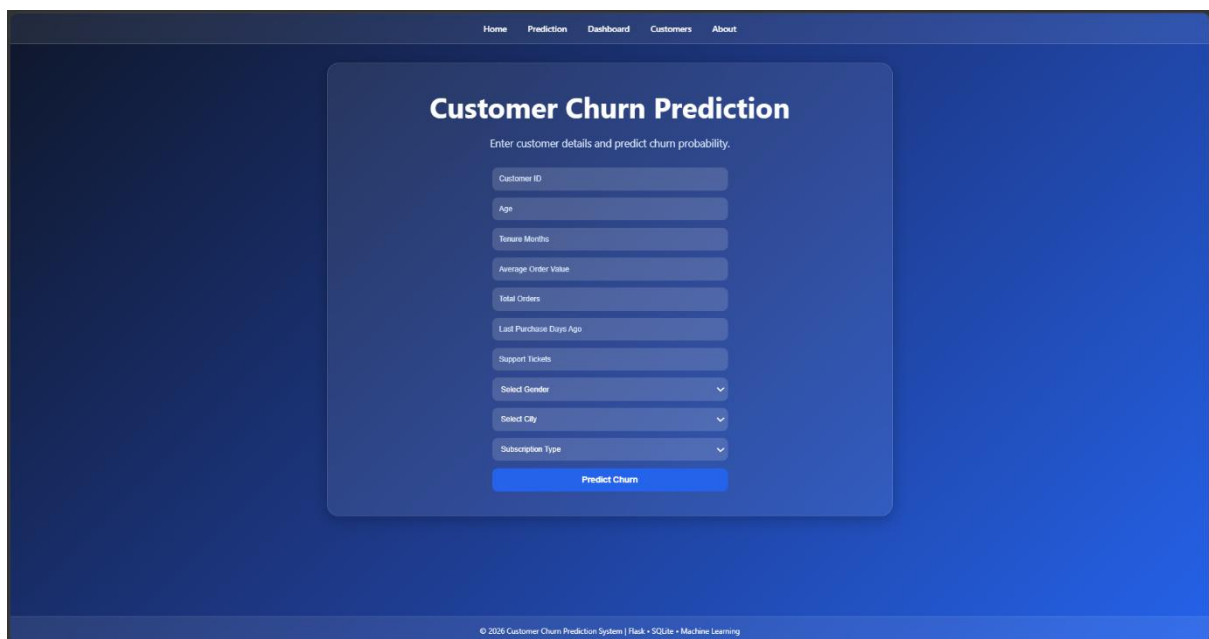
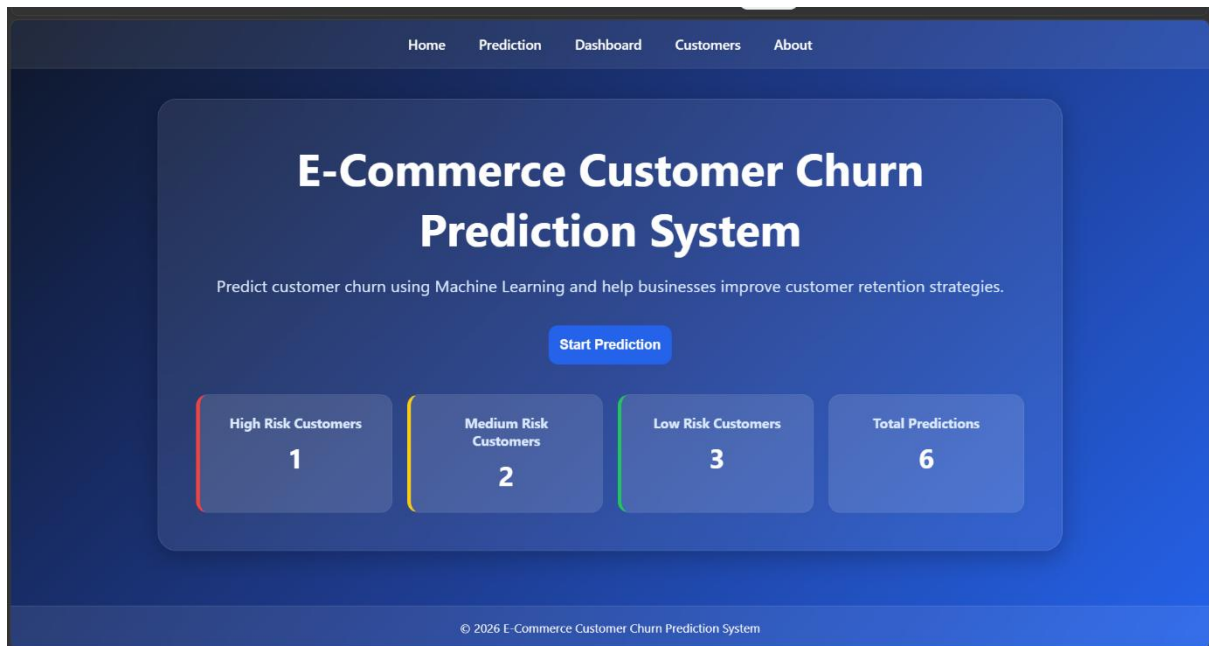
Prediction result visualization.

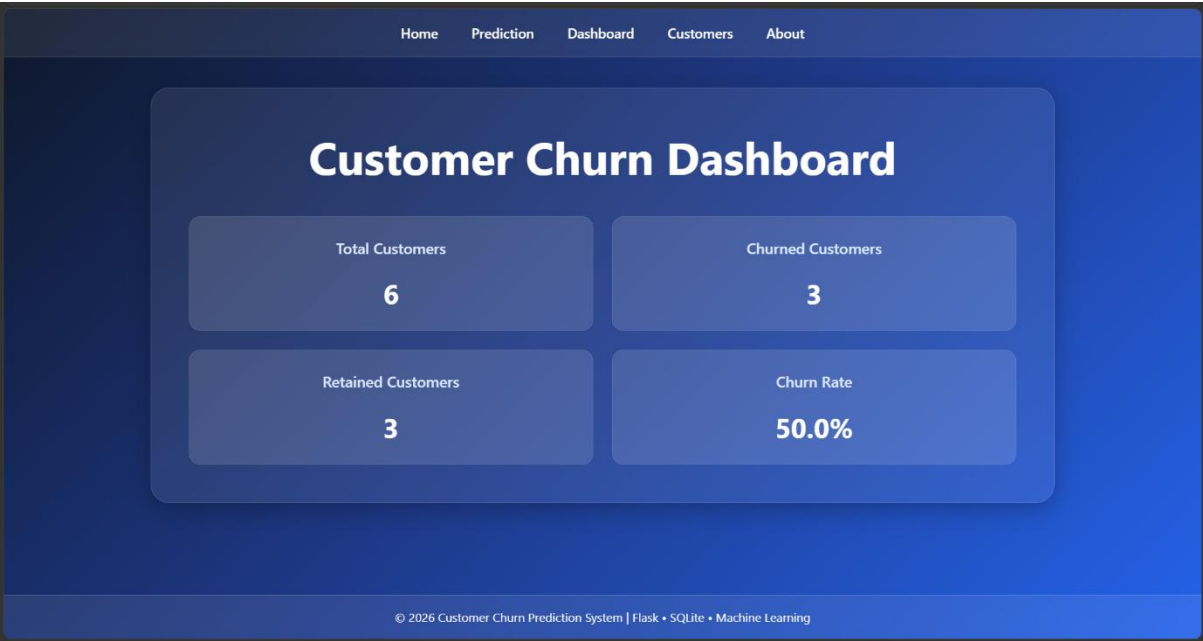
GitHub Repository

GitHub Link:

<https://github.com/NSNihalreddy/VELTECH-SUMMER-INTERNSHIP-/tree/main/Innolift%20Ventures/Customer-Churn-Prediction-System>

Project Screenshots

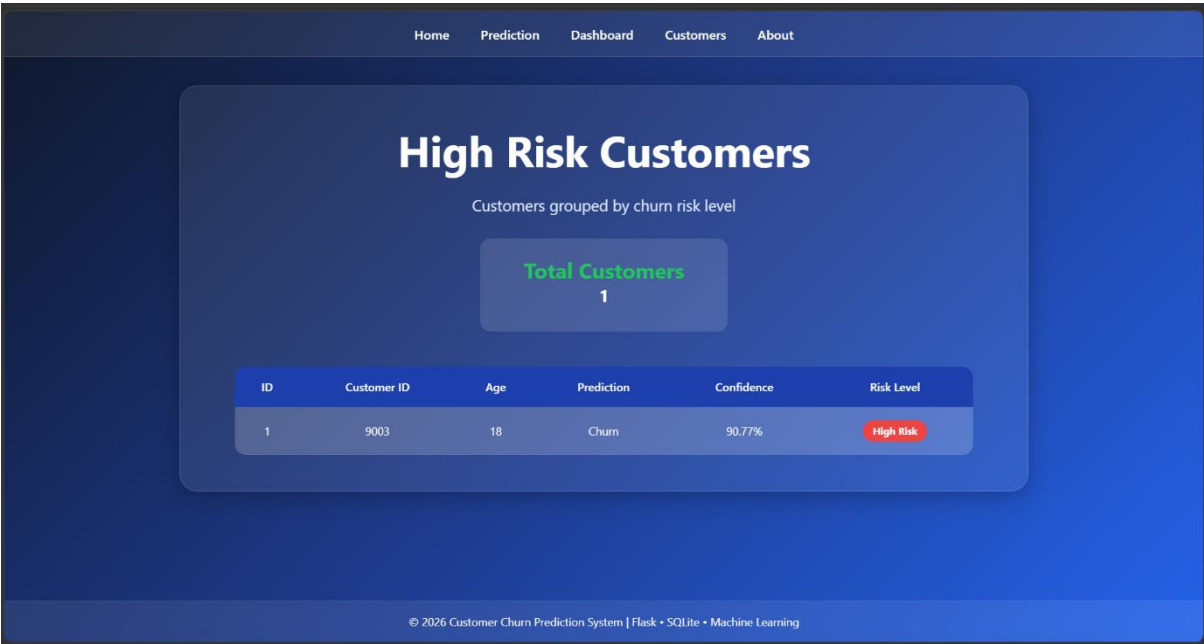
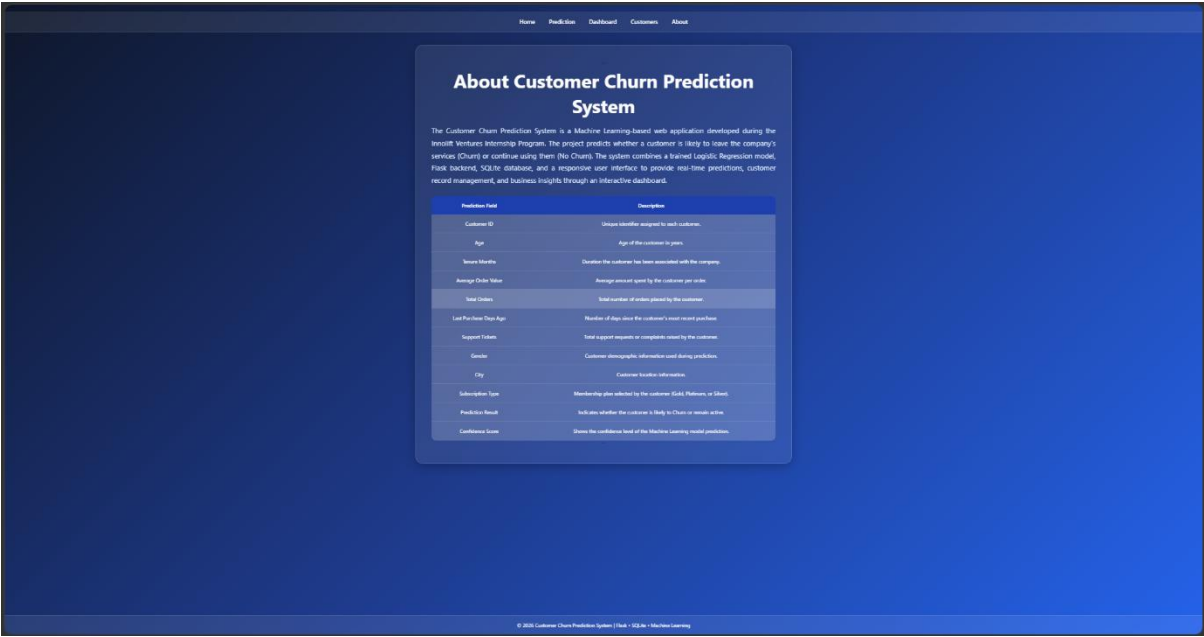


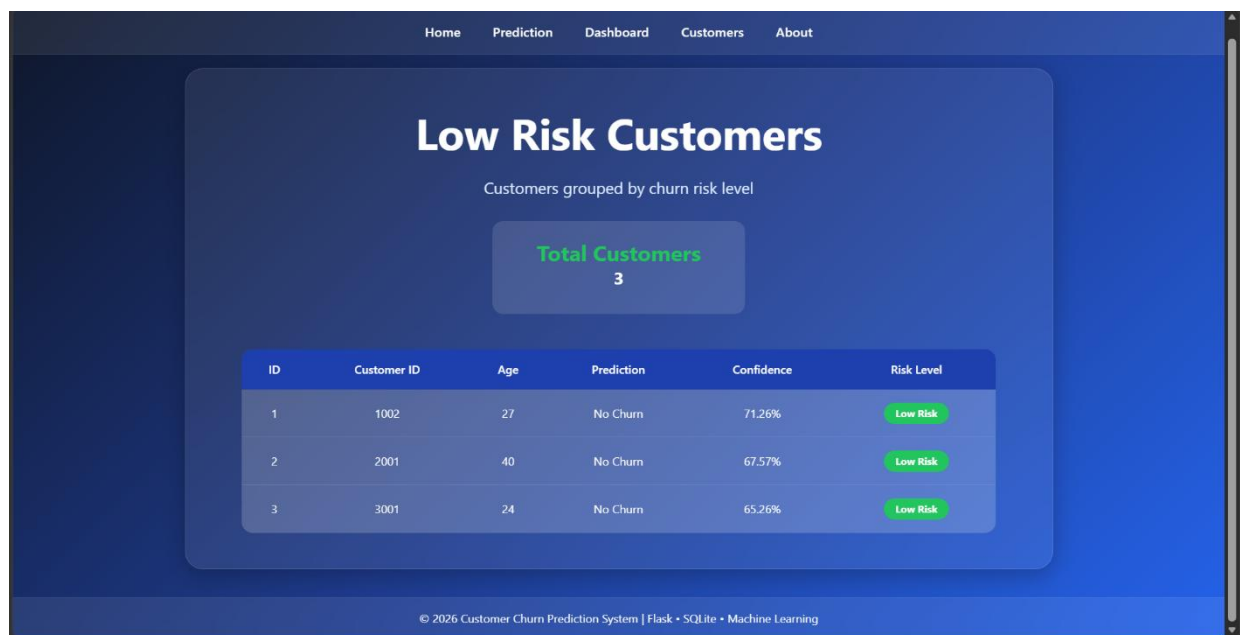
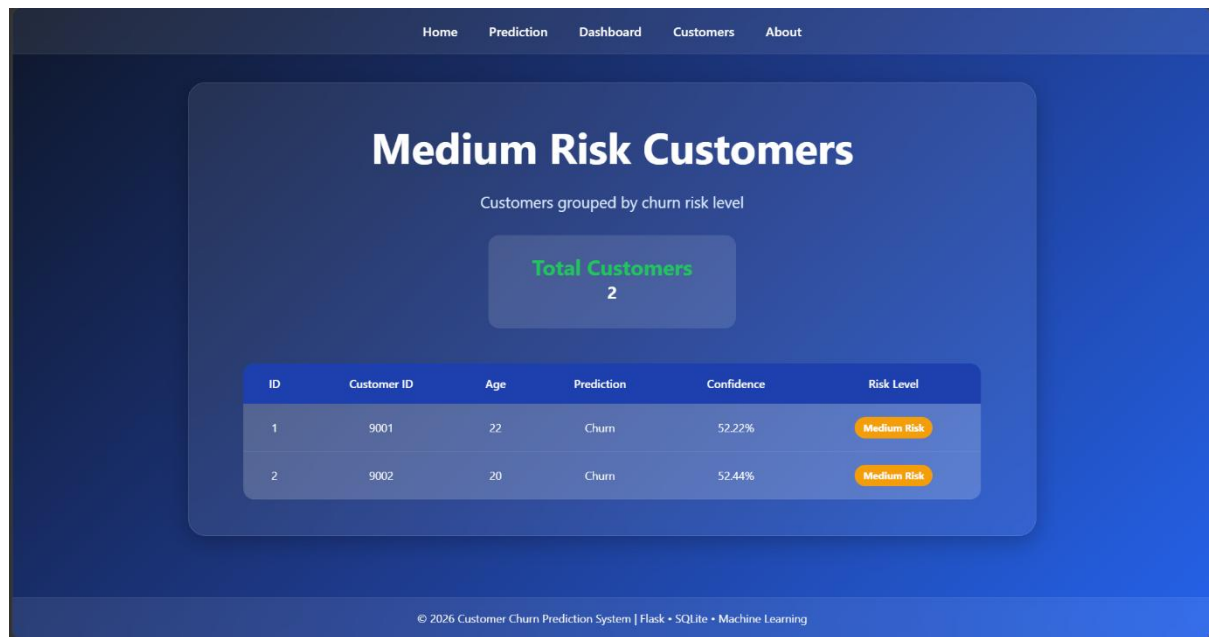


Home Prediction Dashboard Customers About

Customer Records

ID	Customer ID	Age	Tenure	Orders	City	Subscription	Prediction	Confidence	Risk Level	Action
1	1002	27	18	20	Kolkata	Silver	No Churn	71.26%	Low Risk	Delete
2	2001	40	3	2	Kolkata	Silver	No Churn	67.57%	Low Risk	Delete
3	3001	24	2	2	Delhi	Silver	No Churn	65.26%	Low Risk	Delete
4	9001	22	1	1	Delhi	Silver	Churn	52.22%	Medium Risk	Delete
5	9002	20	1	1	Delhi	Silver	Churn	52.44%	Medium Risk	Delete
6	9003	18	1	1	Delhi	Silver	Churn	90.77%	High Risk	Delete





Conclusion :

The E-Commerce Customer Churn Prediction System successfully demonstrates the integration of Machine Learning, Flask, SQLite, and Web Development technologies into a complete business application. The system predicts customer churn, identifies risk levels, and provides retention recommendations that can assist organizations in reducing customer attrition and improving customer satisfaction. This project showcases end-to-end machine learning

deployment and full-stack application development skills acquired during the internship program.